Feature Engineering

Chest X-Ray Multi-Condition Detection — DataML Challenge

1. Why hand-crafted features

The challenge rules do not allow neural networks or pretrained models. So the model cannot learn its own features from raw pixels the way a CNN would. Instead, every feature has to be written by hand using classical image processing (NumPy, OpenCV, scikit-image, SciPy, PyWavelets). Each feature is a formula that turns one X-ray image into a small set of numbers describing one specific property of the image, such as brightness, texture, or shape.

Ten feature families were designed in total, each capturing one specific property of the image. Not all of them are used at once — the notebook groups them into feature sets of increasing size (A through E), and the current run uses feature set C: intensity, spatial, texture, gradient, morphology, and symmetry. Together these six families produce 134 raw numeric features per X-ray image.

2. Feature families used (feature set C)

Every family is computed on the same preprocessed image, so the comparison between families is fair. The table below summarizes each family, including how many individual features it contributes.

Family	# Features	What it measures	Why included	Disease sign it targets
Intensity	31	Overall brightness of the image: global stats, percentiles, and a 16-bin brightness histogram.	Diseased lung tissue often looks denser (whiter) or hazier than normal air-filled lung, so overall brightness statistics can shift.	Infiltration, Effusion
Spatial grid	48	The image split into a 3x3 grid and into top/middle/bottom zones; brightness stats per cell/zone.	Tells the model where in the chest something looks unusual, not just whether the image as a whole looks unusual.	Effusion (lower zone), Pneumothorax (upper zone)
Texture (LBP + GLCM)	30	Local pattern of the image: whether nearby pixels are smooth, patchy, or repeating in a fixed way.	Some conditions create patchy or irregular tissue patterns instead of a smooth lung field.	Infiltration, Nodule
Gradient / edge	12	How sharp and in what direction intensity changes are (edges), plus overall edge density.	Some conditions produce a sharp line or boundary that a normal lung field does not have.	Pneumothorax, Nodule
Morphology (shape)	8	Shape statistics of bright/dark regions: count, size, roundness, solidity.	A round, solid, well-defined shape is a basic sign of a nodule;	Nodule

Family	# Features	What it measures	Why included	Disease sign it targets
			morphology tries to capture that shape directly.	
Symmetry	5	Compares the left half of the chest to the right half (brightness, histogram shape, edge density).	The two lungs are normally close to mirror images of each other; many conditions affect only one side.	Pneumothorax, Effusion
Total (feature set C)	**134**	Combined raw feature count extracted per image, before cleaning.		

3. Full list of features, by family

This is the detailed breakdown behind the counts in Section 2 — the actual column names produced by extract_all_features() for feature set C, grouped by what they describe. Numbered suffixes (e.g. _0, _1, …) index histogram bins, grid cells, or GLCM angle/distance combinations rather than being separate hand-picked features.

A. Intensity (31 features)

Global distribution stats (9):

`int_mean, int_median, int_std, int_var, int_min, int_max, int_skew, int_kurt, int_entropy`

Percentiles (6):

`int_p5, int_p25, int_p50, int_p75, int_p95, int_iqr`

Brightness histogram (16): *(16-bin histogram of pixel intensities)*

`int_hist_0 … int_hist_15`

B. Spatial grid (48 features)

3×3 grid cells (36): *(row, col each 0–2 → 9 cells × 4 stats)*

`grid_{row}_{col}_mean, grid_{row}_{col}_std, grid_{row}_{col}_p75, grid_{row}_{col}_entropy`

Top / middle / bottom zones (12):

`zone_top_mean, zone_top_std, zone_top_p75, zone_top_entropy, zone_mid_(…), zone_bottom_(…)`

C. Texture (LBP + GLCM) (30 features)

Local Binary Pattern histogram (18): *(uniform LBP, 16 sample points → 18-bin histogram)*

`lbp_s1_0 … lbp_s1_17`

Gray-Level Co-occurrence Matrix stats (12):

`glcm_s1_contrast_mean, glcm_s1_contrast_std, glcm_s1_dissimilarity_mean, glcm_s1_dissimilarity_std, glcm_s1_homogeneity_mean, glcm_s1_homogeneity_std, glcm_s1_energy_mean, glcm_s1_energy_std, glcm_s1_correlation_mean, glcm_s1_correlation_std, glcm_s1_ASM_mean, glcm_s1_ASM_std`

D. Gradient / edge (12 features)

Gradient magnitude & edge density (4):

`grad_mag_mean, grad_mag_std, grad_mag_p95, edge_density`

Gradient orientation histogram (8):

`grad_orient_hist_0 … grad_orient_hist_7`

E. Morphology (shape) (8 features)

Connected-component shape stats (6):

`morph_n_components, morph_area_mean, morph_area_max, morph_circularity_mean, morph_eccentricity_mean, morph_solidity_mean`

Top-hat / black-hat filtering (2):

`morph_tophat_mean, morph_blackhat_mean`

F. Symmetry (5 features)

Left–right comparison (5):

`sym_intensity_mean_diff, sym_intensity_abs_diff, sym_intensity_std_diff, sym_hist_divergence, sym_edge_density_diff`

134 raw features in total (31 + 48 + 30 + 12 + 8 + 5) are extracted per image under feature set C. This is the number produced before any cleaning is applied — see Section 6 for what happens to that count next.

4. Two versions of the image feed the features

Preprocessing produces two aligned versions of each X-ray:

- base — normalized only. Used for the plain brightness and region statistics (intensity, spatial grid, symmetry), because CLAHE contrast enhancement would distort raw brightness comparisons.
- texture — CLAHE contrast enhancement plus gamma correction. Used for texture, edge, and shape families, because these depend on local contrast being visible, which CLAHE improves.

Using the right version for the right family matters: applying CLAHE before measuring raw brightness would make the intensity features describe the contrast enhancement instead of the actual X-ray.

5. Why a lung-area mask was added

A chest X-ray contains the lungs, the rib cage, the spine, and black background from the scan. Bones and background do not carry disease information but still affect an average taken over the whole image. A lung-area-restricted feature family (part of the larger feature sets D/E, not used in the current feature set C run) segments an approximate lung area using thresholding and shape operations, then repeats intensity and left/right comparisons using only pixels inside that area. A segmentation-quality flag is stored alongside these features so that unreliable segmentations can be identified rather than silently trusted.

6. How the features connect to the spatial grid

The spatial grid family is what lets location matter to the model. Without it, two X-rays with the same overall brightness would look identical to the model even if one has a problem confined to the top of one lung and the other is uniformly hazy. Splitting the image into a grid and into top/middle/bottom zones lets the same intensity-style statistics be computed per region instead of only once for the whole image.

7. Cleaning and selecting features before modelling

Raw feature extraction produces 134 columns for feature set. Two cleanup steps are applied before any model sees them:

- Feature cleaning (unconditional, no label used): near-constant columns are dropped (variance threshold), and when two columns are highly correlated with each other (> 0.95), one of the pair is dropped, since it adds

redundancy rather than new information. The exact final column count depends on this dataset's actual correlations — it is printed by clean_features() as X_train.shape[1] and will typically be lower than 134.

- Feature selection (inside the training fold only, using mutual information): for each condition separately, only the top-k features most informative for that specific condition are kept before the model is scaled and trained. This step is placed inside the cross-validation pipeline so it never sees validation or test data.

8. How feature usefulness was checked

The notebook includes a few diagnostic checks to see whether these feature families were actually behaving as intended, rather than assuming it:

- Mutual information ranking: for each condition, mutual information is computed between every feature and the label, then either averaged within each family (radar chart) or ranked individually (bar charts of the top 10 features per condition, shown below).
- PCA 2D projection: all cleaned features are reduced to two dimensions and plotted per condition, to see visually whether positive cases form a separate cluster from normal scans, or whether they overlap heavily.
- Left–right symmetry check: a separate diagnostic compares the average left/right brightness difference for images with a condition versus normal images, with sample size and standard error shown, to check whether the symmetry family is picking up a real signal before trusting it inside the model.

These checks are diagnostic, not proof of causation — a feature scoring high mutual information for a condition means it is statistically useful for telling that condition apart from normal in this dataset, not that it captures the true clinical cause.

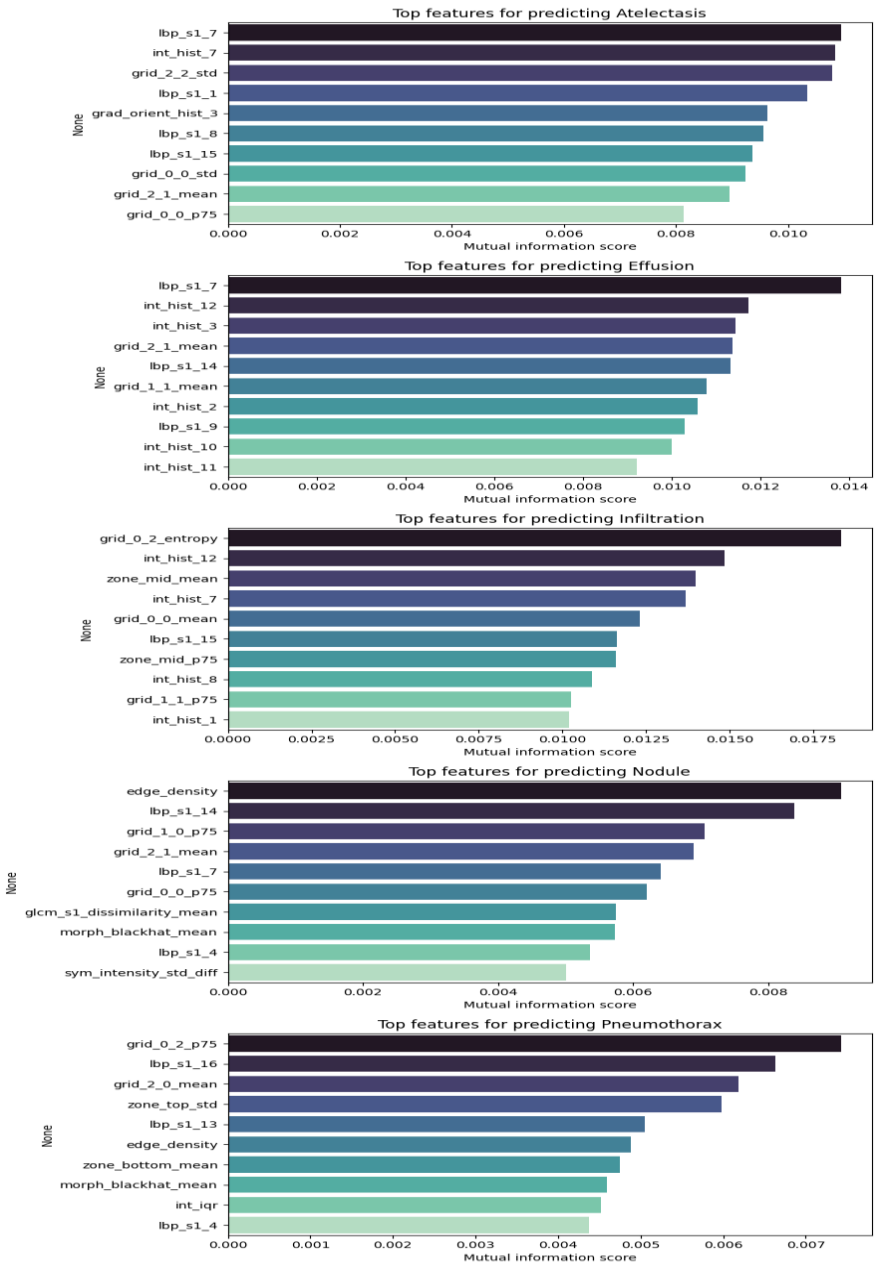


Figure 1. Top 10 individual features by mutual information score, per condition (Atelectasis, Effusion, Infiltration, Nodule, Pneumothorax). Note that texture (lbp_, glcm_) and intensity-histogram features dominate most conditions, while Nodule and Pneumothorax lean more on edge/gradient and grid-location features — consistent with the family-level pattern in Section 2.

9. Limitations to keep in mind

- Feature-to-condition mapping in the Section 2 table (for example, symmetry → pneumothorax) is a reasonable expectation based on how these conditions typically present, not a guarantee — the mutual information and PCA diagnostics are what actually tests it on this dataset.
- The 134-feature count is the raw count for feature set only; it is not the number of features the models are ultimately trained on, since clean_features() and the per-label mutual-information selector both reduce it further downstream.
- Classical hand-crafted features are inherently more limited than what a CNN could learn directly from pixels; this is a constraint of the challenge rules, not a design choice.